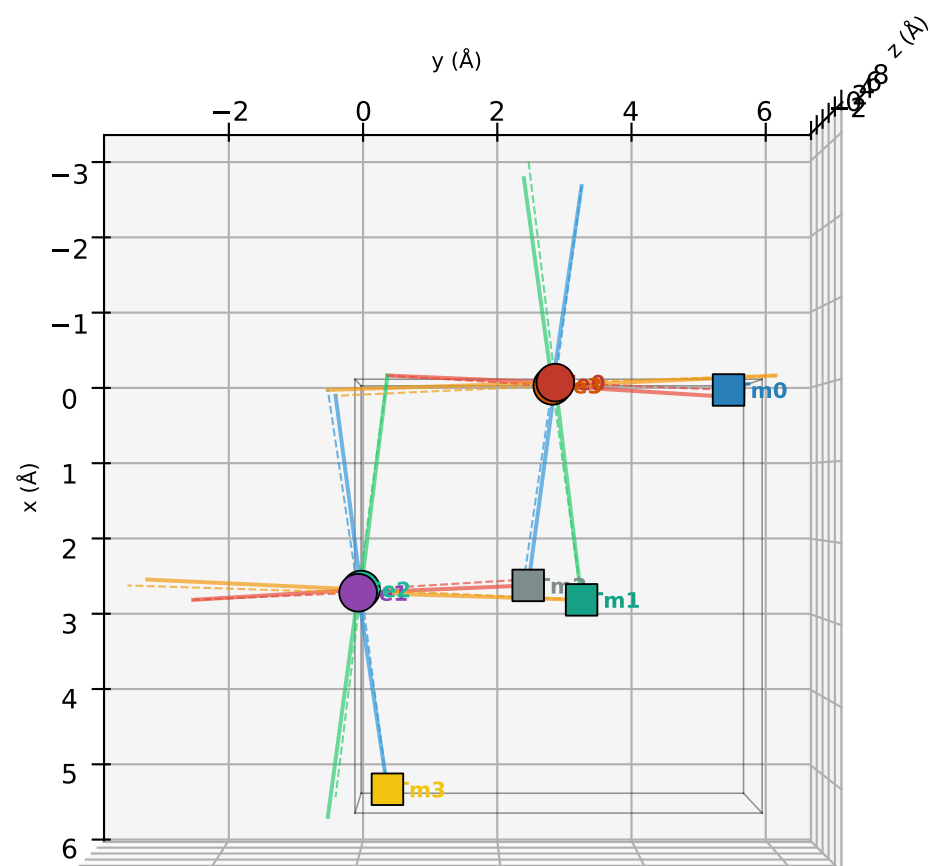
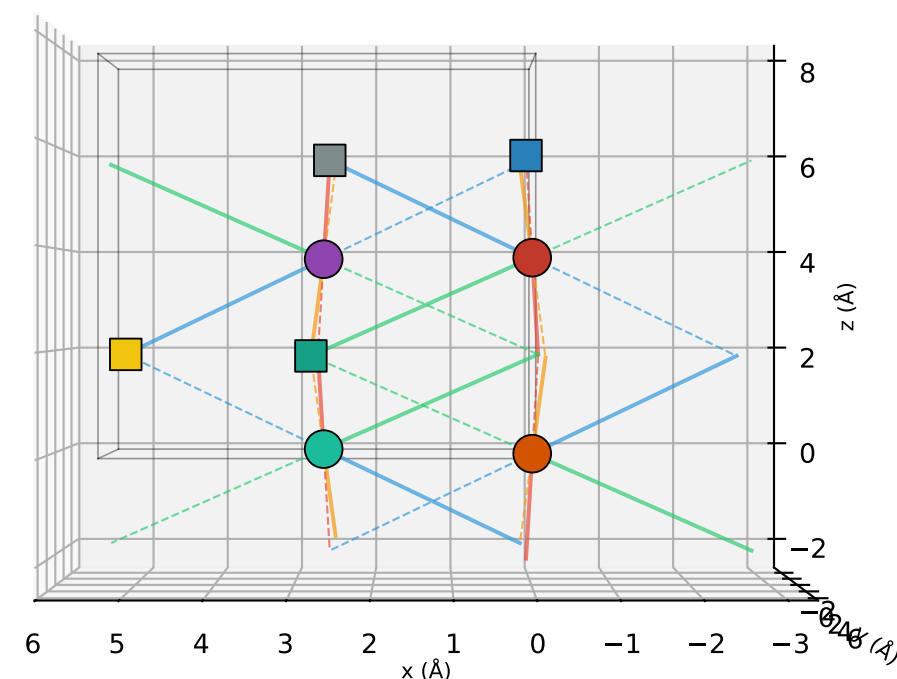


TmFeO3 crystal structure — Fe-Tm chi coupling bonds (Pbnm Γ_2 phase, 4 orbits $\times$ 2 bonds per orbit = 8 unique bond types)

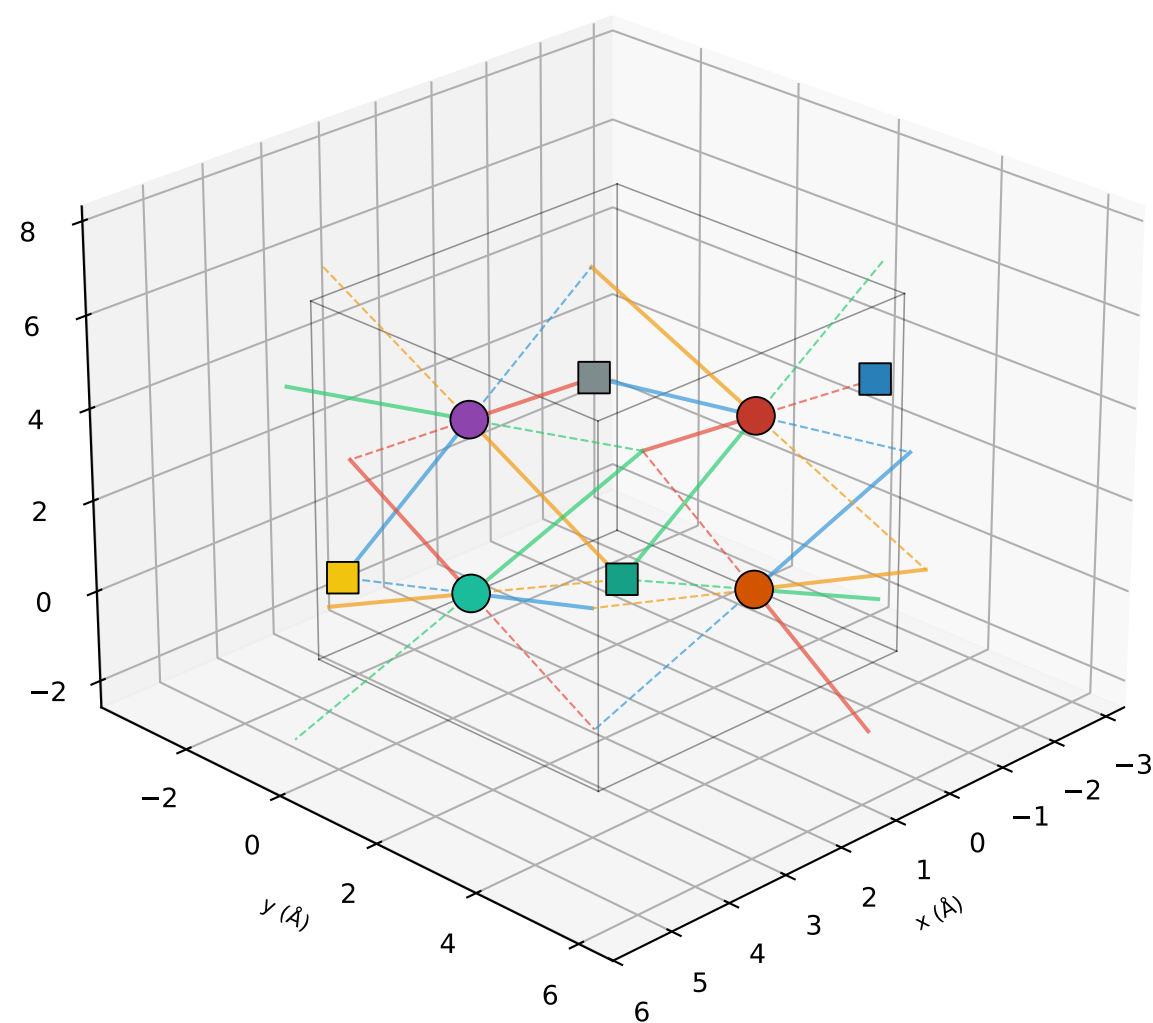
Looking down c-axis (a-b plane)



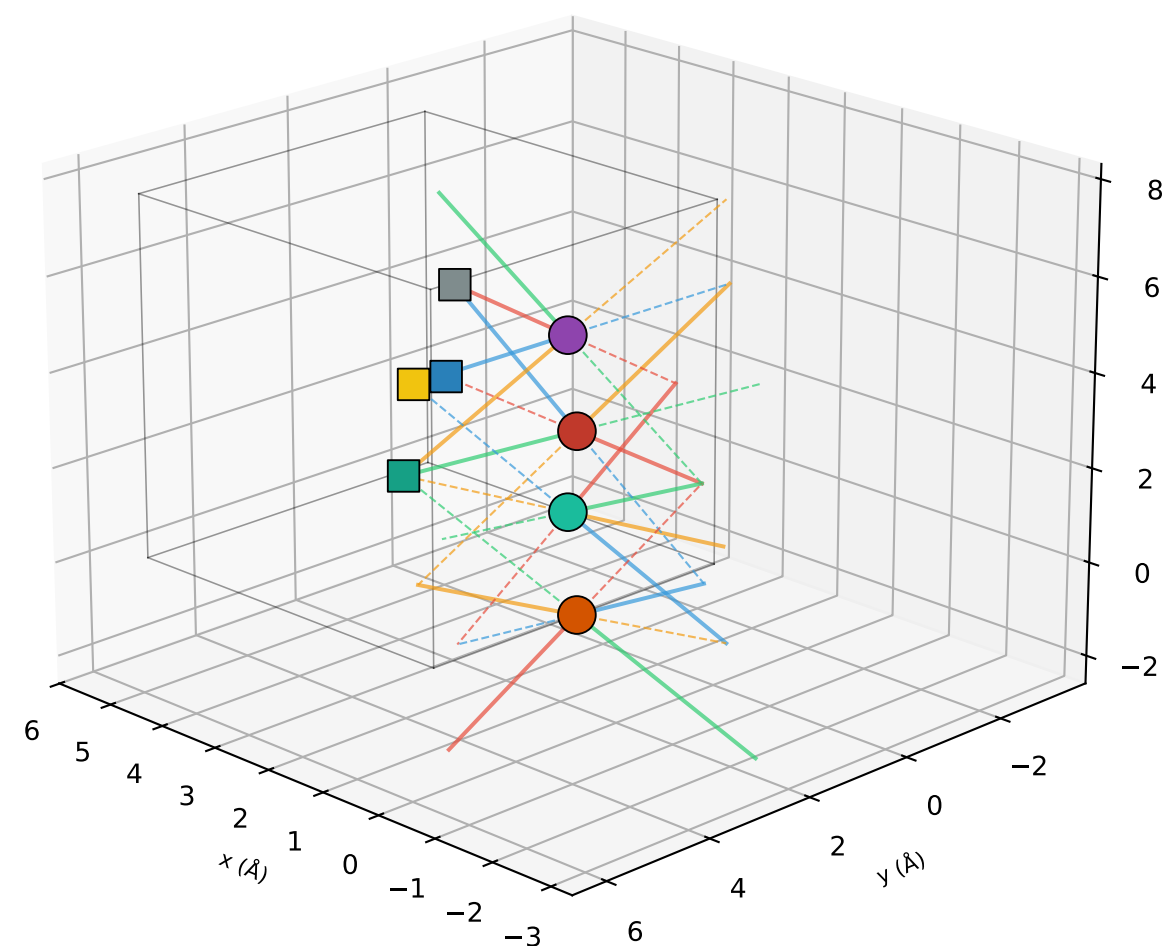
Looking down b-axis (a-c plane)



Perspective 1 (ab-plane tilt)



Perspective 2 (bc-plane tilt)



Orbit 1 Orbit 2 Orbit 3 Orbit 4 — even bond (sign₅₇=+1) ---- odd bond (sign₅₇=-1)